

WILLAMETTE VALLEY FOODS, INC.

FOOD SAFETY PLAN

Ready-to-eat hummus and bean dips

Salem Plant, 1850 Fairgrounds Road NE, Salem, Oregon 97301

Effective date: May 12, 2025

Prepared by: Dana Whitfield, Quality Manager, Preventive Controls Qualified Individual

(PCQI certificate, FSPCA course, March 2024)

Approved by: Plant Manager

1. Facility and product description

Willamette Valley Foods produces refrigerated ready-to-eat hummus and bean dips in 8-ounce and 16-ounce sealed plastic tubs for retail sale in Oregon, Washington and Idaho. The Salem plant operates one shift, five days a week, with 28 production employees. Products are made from cooked chickpeas or black beans, tahini, lemon juice, garlic, olive oil, salt and spices, blended, filled, sealed, coded and held refrigerated at or below 41°F. Shelf life is 45 days refrigerated. Products are labeled with the allergen statement "Contains: sesame" for all hummus items.

2. Process flow

Receiving of dry beans, tahini and other ingredients → dry storage or refrigerated storage → soaking and cooking of beans (kettle, 212°F, 60 minutes minimum) → cooling in the blast chiller → blending with other ingredients → filling and sealing → date coding → metal detection → refrigerated storage → shipping.

3. Hazard analysis

Step	Hazard	Type	Requires a preventive control?	Justification
Receiving tahini	Sesame allergen	Allergen	Yes	Undeclared allergen if mislabeled; cross-contact with bean dips made without tahini
Receiving dry beans	Foreign material (stones, metal)	Physical	Yes	Field debris in dry beans, controlled at metal detection and by supplier screening
Cooking of beans	Survival of vegetative pathogens (Salmonella, Listeria monocytogenes)	Biological	Yes	Cook step is the kill step for the finished product
Cooling	Growth of Clostridium perfringens and Bacillus cereus	Biological	Yes	Slow cooling of a cooked legume allows spore-former outgrowth
Blending and filling	Listeria monocytogenes from the environment	Biological	Yes	RTE product exposed to the environment after the kill step
Refrigerated storage	Growth of Listeria monocytogenes	Biological	Yes	Refrigerated RTE product with a 45-day shelf life
Date coding	None significant	—	No	—
Metal detection	Metal fragments	Physical	Yes	Final screen for metal from equipment and ingredients

4. Preventive controls

Hazard	Preventive control	Type	Critical limit or parameter
Vegetative pathogens at cook	Cook beans in the kettle to an internal temperature of at least 212°F for at least 60 minutes	Process	Minimum 212°F for 60 minutes, measured with a calibrated probe thermometer
Spore-former growth at cooling	Cool cooked beans from 135°F to 70°F within 2 hours and to 41°F within a further 4 hours in the blast chiller	Process	135°F to 70°F in 2 hours; 70°F to 41°F in 4 hours
Listeria in the RTE environment	Sanitation of food-contact surfaces before each production run and between products; environmental monitoring swabs weekly	Sanitation	ATP swab pass before start-up; Listeria spp. swabs, 10 sites weekly, zone 1 and 2
Sesame allergen	Allergen labeling review for every label at receipt and at printing; product scheduling with hummus items last in the day, followed by full washdown	Allergen	Label matches the formulation sheet; washdown verified by ATP swab before the next non-sesame product
Metal	Metal detection of every sealed tub	Process	Reject at 2.0 mm ferrous, 2.5 mm non-ferrous, 3.0 mm stainless test pieces

5. Monitoring

The kettle operator records the internal temperature and the time at the start and end of each cook batch on the cook log. The blast chiller operator records the temperature at 2 hours and at 6 hours on the cooling log. The sanitation lead records the ATP swab result before each start-up. The line lead runs the three metal test pieces through the detector at start-up, every two hours, and at the end of the run, and records the result. The quality technician reviews every label lot against the formulation sheet at receipt and signs the label log.

6. Corrective actions

If a cook batch does not reach 212°F for 60 minutes, the batch is re-cooked to the critical limit or discarded, and the cause is recorded. If cooling exceeds the time limits, the batch is discarded. If an ATP swab fails, the surface is re-cleaned and re-swabbed before production starts. If a *Listeria* spp. swab is positive, the site is cleaned and re-swabbed, adjacent sites are added, and product made on that line since the last negative swab is held pending investigation. If the metal detector fails a test piece, all product since the last passing test is re-run through the detector. If a label does not match the formulation, the lot is quarantined and not used. All corrective actions are recorded on the corrective action log and reviewed by the PCQI within seven days.

7. Verification

The PCQI reviews the cook, cooling, sanitation, allergen and metal detection records within seven working days of their creation. Probe thermometers are calibrated in ice water and boiling water weekly, and the metal detector is serviced by the manufacturer annually. Finished product is tested for *Listeria monocytogenes* monthly by an accredited laboratory. Environmental monitoring results are trended quarterly.

8. Supply-chain program

Tahini and dry beans are received only from approved suppliers who provide a current third-party audit certificate and a certificate of analysis for each lot. The approved supplier list is maintained by the quality manager.

9. Records

All records named in this plan are kept at the Salem plant for at least two years from the date of the record and are available to FDA and the Oregon Department of Agriculture on request. Records are reviewed and signed by the PCQI as described in section 7.

10. Training

All production employees receive food hygiene training at hire and annually. The PCQI holds a current FSPCA certificate. Sanitation and line leads are trained on the monitoring procedures in this plan.